



Experiment No. 2

Title: To study of GSM architecture & signal techniques

Problem Statement: To study of GSM architecture & signal techniques.

Objective: To understand the concept of hardware description language.

Theory:

A GSM network comprises of many functional units. These function and interfaces are explained in this. GSM network can be divided into-

- Mobile Station (MS)
- Base Station Subsystem (BSS)
- Network Switching Subsystem (NSS)
- Operation Support Subsystem (OSS)



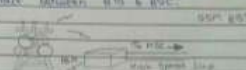
GSM - The Mobile Station

The MS consists of physical equipment, such as the radio transceiver, display & digital signal processor, and SIM card. It provides the air interface to user in GSM networks. In such, other services are also provided, which includes:

- Voice telephony
- Data transfer services
- The features supplementary services

The MS also provides the receiver for SMS messages enabling user to toggle with the voice & data user. The mobile location process to voice message system. The MS also provides access to various other services.

The BTS and BSC communicates across specified radio interface, enabling operation between component that are made by different suppliers. The radio components of BSS may consist of 4 to 16 BTS cells. BSS may have 1 or more BS. BSS uses radio interface between BTS & BSC.



BTS - BTS houses radio transceiver that define a cell and handles the radio link protocols with MS.



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- Data bearer services
- The features' supplementary services

The MS also provides the receptor for SMS message enabling user to toggle betⁿ the voice & data user. The mobile facilities access to voice message system. The MS also provide access to various data service available in GSM network.

- X.25 packet switching through a synchronous or asynchronous connection.
- GPRS using X.25 or IP based data transfer method at the speed of 115 Kbps.

SIM - Provides personal so user can have access to all services irrespective of location and use of specific terminal. You need to insert SIM into GSM cellular phone to receive calls at phone.

GSM - The Base Station Subsystem (BSS)

The BSS is composed of 2 parts -

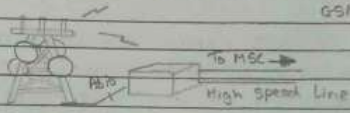
- Base Transceiver Station (BTS)
- Base Station Controller (BSC)

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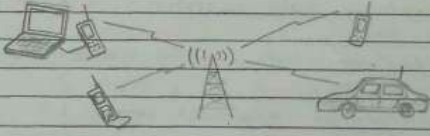
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The BTS and BSC communicates across specified Abis interface, enabling operation between component that are made by different suppliers. The radio components of BSS may consists of 4 to 700 cells. BSS may have 1 or more BS. BSS uses Abis interface between BTS & BSC.

GSM BSS



BTS- BTS houses radio transceivers that define a cell and handles the radio link protocols with MS. In large area, a no. of BTSs are deployed.



GSM BTS

It corresponds to transceivers & antennas used in each cell of network. A BTS is usually placed in center of cell. Its transmitting power defines the size of a cell. Each BTS has between 1 to 16 transceivers, depending on density of users in cell. It includes following functions-

- Encoding, encrypting, multiplexing, modulating & feeding the RF signals to the antenna.
- Transcoding & rate adaptation.





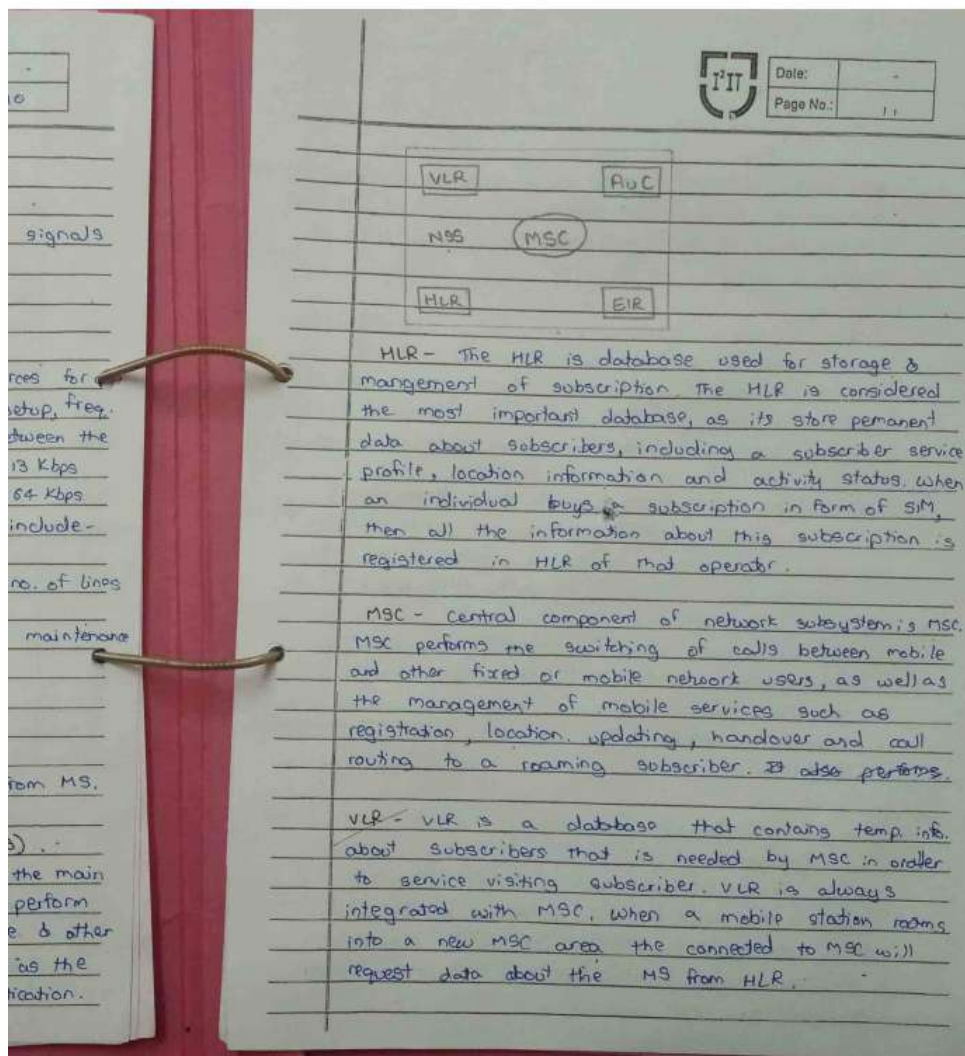
- Time to frequency synchronization.
- Voice through full or half rate service.
- Decoding, decrypting and equalizing signals.
- Random access detection.
- Timing advances.
- Uplink channel measurements.

BSC - The BSC manage the radio resources for one or more BTS. It handles radio channel setup, freq. hopping, handovers. It is the connection between the mobile and the MSC. BSC also translates 13 Kbps voice channel used over the radio link to 64 Kbps channel used by PSDN. The functions include-

- Control of frequency hopping.
- Performing traffic concentration to reduce no. of lines from MSC.
- Providing an interface to the operation and maintenance center for BSS.
- Reallocation of frequencies among BTS.
- Time & frequency synchronization.
- Power management.
- Time-delay measurement of received signal from MS.

GSM - Network Switching Subsystem (NSS).

The Network switching system (NSS), the main part of which is mobile switching center, perform the switching of calls between the mobile & other fixed or mobile network users, as well as the management of mobile service such as authentication.





AUC - Authentication Centre is a protected database that stores a copy of secret key stored in SIM card, which is used for authentication & ciphering of radio channel.

EIR - Equipment Identity Register is a database that contains a list of all valid mobile equipment on the network, where IMEI identifies each MS. An IMEI is marked as invalid if it has been reported stolen or is not type approved.

GSM - The Operational Support Subsystem (OSS)

The operation and maintenance center is connected to all equipment in switching system & BSC. The implementation of omc is called the operation & support system.

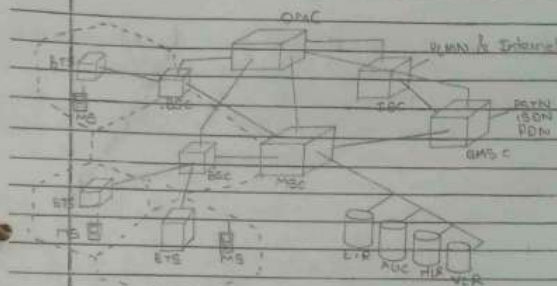
Here are some of omc functions.

- Administration and commercial operation
- Security Management
- Network configuration, operation & Performance Management
- Maintenance Tasks.

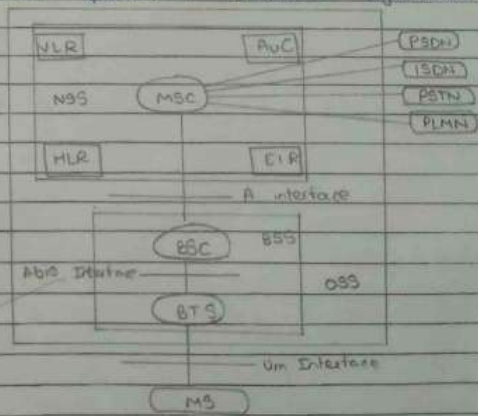
The operation & maintenance functions are based on concepts of TMN. The OSS is functional entity from which network operator monitors & controls the system. The purpose of OSS is to offer the customer cost-effective support for regional & local operation & maintenance activities that are required for GSM network.



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An important function of OSS is to provide a network overview & support the maintenance activities of different operation & maintenance organisation.

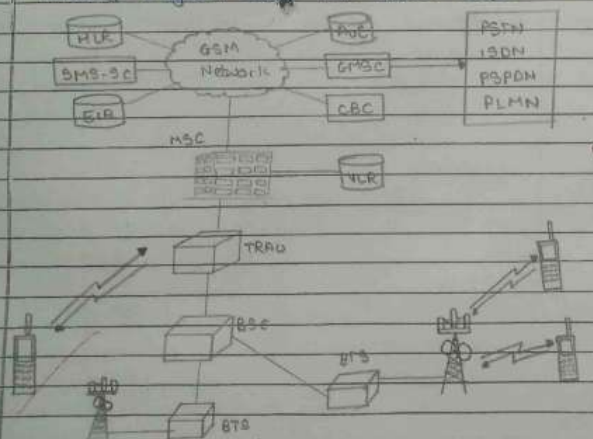




Additional components of GSM architecture comprise of database and messaging system functions.

- Home Location Register (HLR)
- Visitor Location Register (VLR)
- Equipment Identity Register (EIR)
- Authentication Center (AUC)
- SMS Serving center (SMS SC)
- Gateway MSC (GMSC)
- Chargeback Center (CBC)
- Transcoder & Adaptation Unit (TRAU)

Following diagram shows GSM network along with elements





GSM Network Areas

In GSM network, the foll. areas are defined.

- **Cell** - It is a basic service area, one BTS covers one cell. Each cell is given a cell global identity (CGI).
- **Location Area** - A group of cells from LA. MS is area that is paged when a sub. get an call. Each LA is assigned LAI. Each LA is served by one or more BSC.
- **MSC/VLR Service Area** - The area covered by one MSC is called the MSC/VLR service area.
- **PLMN** - The area covered by one network operator is called Public Land Mobile Network (PLMN). Can contain one or more MSCs.

Modulation - It is a process of transforming input data into a format for transmission medium. The transmitted data is demodulated back to original form at receiving end. GSM uses GMSK modulation method.

Access Method - Radio spectrum being a limited resource that is consumed and divided among users, GSM devised a combⁿ of TDMA/FDMA as method to divide bandwidth among users. FDMA divides the freq. of total 15MHz bandwidth into 124 carrier frequencies of 300KHz bandwidth. Each BS is assigned with multiple frequencies & each of freq. is divided into eight timeslots using a TDMA scheme. Each of those slots are used for transmission and reception of data.